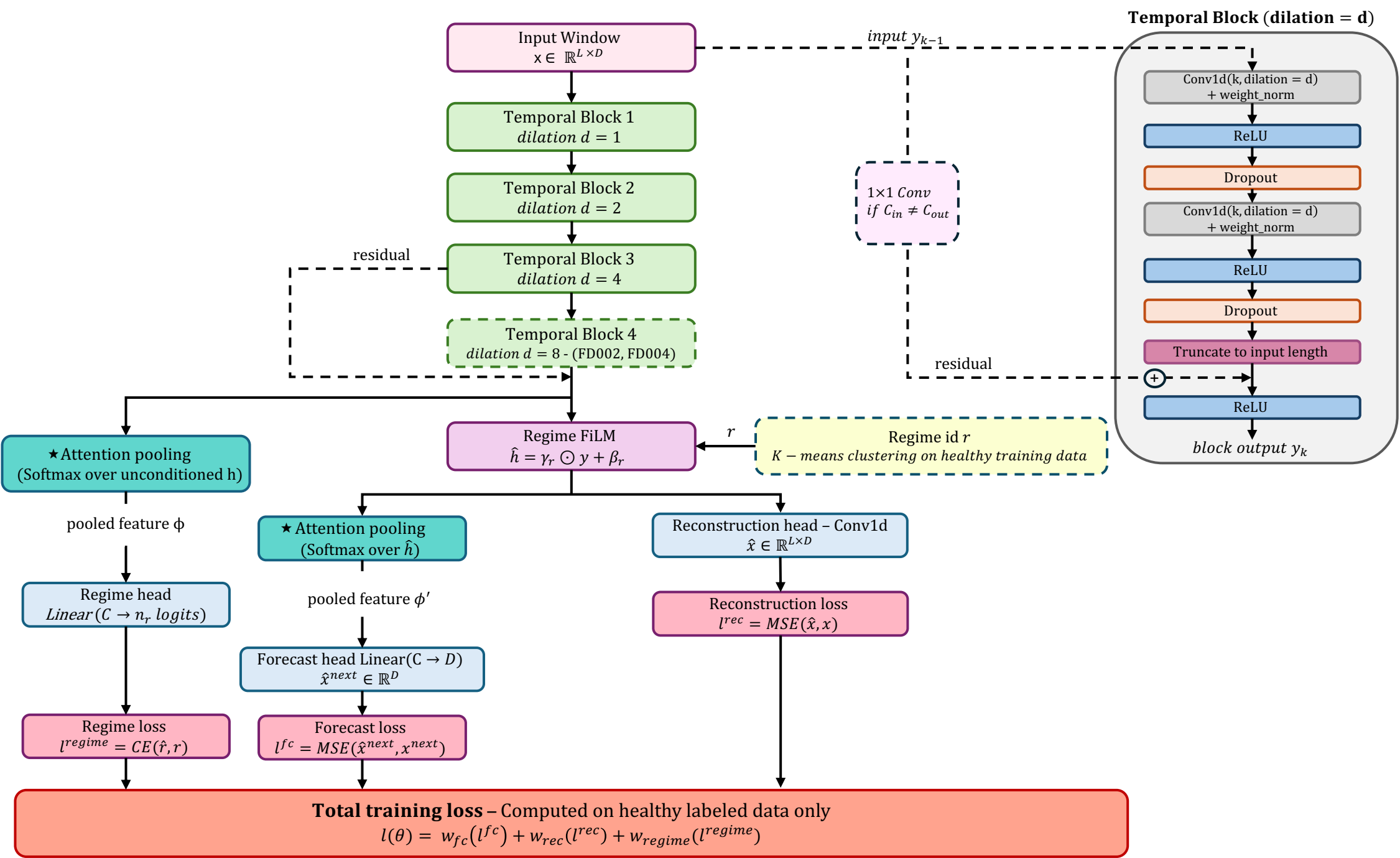

Input Window
x ∈ $\mathbb{R}^{L\times D}$
Temporal Block 1
dilation $d = 1$
Temporal Block 2
dilation $d = 2$
Temporal Block 3
dilation $d = 4$
Temporal Block 4
dilation $d = 8$ - (FD002, FD004)
residual
input y_{k-1}
1×1 Conv
if $C_{in} \neq C_{out}$
residual
Temporal Block (dilation = d)
Conv1d(k, dilation = d)
+ weight_norm
ReLU
Dropout
Conv1d(k, dilation = d)
+ weight_norm
ReLU
Dropout
Truncate to input length
ReLU
block output y_k
★Attention pooling
(Softmax over unconditioned h)
pooled feature ϕ
Regime head
Linear ($C \rightarrow n_r$ logits)
Regime loss
$l^{regime} = CE(\hat{r}, r)$
Regime FiLM
$\hat{h} = \gamma_r \odot y + \beta_r$
r
Regime id r
K − means clustering on healthy training data
★ Attention pooling
(Softmax over $\hat{h}$)
pooled feature ϕ'
Forecast head Linear(C → D)
$\hat{x}^{next} \in \mathbb{R}^D$
Forecast loss
$l^{fc} = MSE(\hat{x}^{next}, x^{next})$
Reconstruction head – Conv1d
$\hat{x} \in \mathbb{R}^{L\times D}$
Reconstruction loss
$l^{rec} = MSE(\hat{x}, x)$
Total training loss – Computed on healthy labeled data only
$l(\theta) = w_{fc}(l^{fc}) + w_{rec}(l^{rec}) + w_{regime}(l^{regime})$